



NCCER Module 22103



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Identification of Heavy Equipment





Section One

Construction and Mining Equipment

Objective

1. Identify and describe heavy equipment used in construction and mining.
 - a. Identify and describe equipment used in excavating and grading construction sites.
 - b. Identify and describe trenching equipment.
 - c. Identify and describe heavy equipment used in mining.



Performance Tasks

Under the supervision of an instructor, trainees should be able to do the following:

1. Identify the various types of heavy equipment and explain their uses.
2. Identify the basic parts of each type of equipment and explain the differences in models among the types of equipment.



1.0.0 Construction and Mining Equipment

- Most construction equipment is designed to perform two or more tasks.



1.1.0 Excavating and Grading Equipment

- Construction tractors
- Dump trucks
- Compactors
- Scrapers
- Backhoe loaders
- Excavators
- Dozers
- Loaders
- Forklifts
- Motor graders



1.1.1 Construction Tractors

- Many tractors are equipped with a power takeoff (PTO) and hydraulic systems can power equipment used for grading, sweeping, or even post-hole digging.



UTILITY TRACTOR



DUAL-WHEEL ARTICULATING TRACTOR



WHEELED ARTICULATING TRACTOR WITH SCRAPERS



RUBBER TRACK TRACTOR

1.1.2 Dump Trucks (1 of 3)

- On-road dump trucks are designed for use on highways and must be registered for that purpose.



(A) TANDEM AXLE DUMPTRUCK



(B) AUXILIARY AXLE DUMPTRUCK

1.1.2 Dump Trucks (2 of 3)

- Off-road trucks are designed for traveling over the rough terrain normally found on a construction site.
- Some off-road trucks are **articulated** so that they can make sharp turns.
- Rigid-frame trucks are used for high-volume operations.
- Crawler dump trucks on rubber tracks are used on wet construction sites.



(A) ARTICULATED TRUCK



(B) RIGID-FRAME DUMPTRUCK



(C) CRAWLER

1.1.2 Dump Trucks (3 of 3)

- A dump truck is usually loaded using an excavator or loader before it hauls material to another location.



1.1.3 Roller/Compactors

Soil must be compacted before structures and roads are built on it.

There are four types of compactors:

- Sheep's foot soil compactor
- Two-drum, steel wheel compactor
- Vibratory steel wheel soil compactor
- **Pneumatic** tire compactor



(A) SHEEP'S FOOT SOIL COMPACTOR



(B) TWO-DRUM, STEEL WHEEL ROLLER (COMPACTOR)



(C) VIBRATORY SOIL COMPACTOR



(D) PNEUMATIC TIRE ROLLER (COMPACTOR)

1.1.4 Scrapers

A scraper is used to remove dirt and other material from a site by scraping it from the surface into the bowl as it moves along.

Four types of scrapers:

- Standard or self-propelled scraper
- Elevating scraper
- Tandem-powered scraper
- Pull-type scraper



(A) SELF-PROPELLED SCRAPER



(B) ELEVATING SCRAPER



(C) TANDEM-POWERED SCRAPER



(D) PULL-TYPE SCRAPER TOWED BY A TRACTOR

1.1.5 Dozers (1 of 2)

- A dozer is a heavy-duty tractor with a pusher blade mounted on the front.
- The blade of a dozer can be hydraulically raised, lowered, or positioned at an angle.
- Most dozers are tracked vehicles, although there are some wheel versions, including heavy-duty four-wheel-drive dozers.
- Dozers are used to clear land, move material, grade, and maintain **haul roads**.



(A) WHEELED DOZER



(B) LOW-TRACK DOZER



(C) HIGH-TRACK DOZER

1.1.5 Dozers (2 of 2)

- Many dozers are equipped with a powered winch.
- Another common attachment is the **ripper**, which is used to break up hard soil, rock, and other materials, such as concrete and asphalt.



(A) WINCH



(B) RIPPER

1.1.6 Loaders

- Loaders are used for loading trucks, stockpiling material, rough grading, digging, and hauling.
- Skid steers are a compact version of the loaders. It is available with a range of attachments.



(A) WHEEL LOADER



(B) CRAWLER LOADER



(C) WHEELED SKID STEER WITH BUCKET



(D) TRACKED SKID STEER WITH AUGER

1.1.7 Forklifts (1 of 3)

- Fixed-mast forklifts have forks that can be raised as high as the limits of the mast allow.
- The mast may be able to tilt forward and backward.



1.1.7 Forklifts (2 of 3)

- Telescoping-boom forklifts can lift material higher than fixed-mast forklifts.
- Some models have a squirt boom, which allows the fork carriage to move horizontally.



1.1.7 Forklifts (3 of 3)

- Articulating forklifts are built with a fork carriage that is hinged so that it pivots left or right of center line.
- They are designed for use in narrow, confined spaces.



1.1.8 Motor Graders (1 of 3)

A motor grader has a centrally located blade, called a moldboard, that can be angled, raised and lowered, and turned horizontally. It can be used to establish a finish grade or for cut and fill work.



1.1.8 Motor Graders (2 of 3)

- Getting the moldboard set to the proper angle is critical.
- GPS devices on the grader control the positioning of the moldboard.
- Motor grader operators must learn how to work with these GPS devices.



(A) MOTOR GRADER WITH ONE GPS ANTENNA



(B) MOTOR GRADER WITH TWO GPS ANTENNAS



(C) GPS BASE STATION WITH ANTENNA AND RADIO RECEIVER

1.1.8 Motor Graders (3 of 3)

- **Scarifiers** rip up the ground so that the moldboard has an easier job of breaking up and moving the dirt.
- A ripper mounted on the rear of a grader breaks up the ground as the grader passes over a given section of ground.



SCARIFIER SHANKS
A SCARIFIER ON THE FRONT OF A MOTOR GRADER



RIPPER SHANKS
A RIPPER ON THE REAR OF A MOTOR GRADER

1.2.0 Trenching and Excavating Equipment

- Excavators and backhoe loaders are commonly used in preparing sites for construction.
- Either of these machines may be used to load soil onto a dump truck.
- They may also dig trenches for the installation of pipelines and utilities.



1.2.1 Backhoe Loaders

- The loader attachment moves material and fills in ditches.
- The backhoe is used to dig and clean ditches, excavate hard-to-reach areas, and hoist light loads.
- Backhoes are equipped with stabilizer legs (outriggers) that keep the backhoe from tipping or rolling, widen the stance of the machine, and increase capacity.



(A) BACKHOE LOADER



(B) TOOL CARRIER



(C) BACKHOE COMPONENTS

1.2.2 Excavators (1 of 3)

The excavator is used for:

- Digging trenches and foundations
- Loading soil, stone, gravel, and other materials into dump trucks and scrapers
- Demolishing structures
- Clearing land
- Lifting and placing tools and materials



(A) HYDRAULIC CRAWLER EXCAVATOR



(B) HYDRAULIC WHEEL EXCAVATOR



(C) COMPACT EXCAVATOR

1.2.2 Excavators (2 of 3)

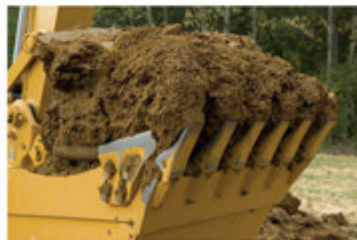
Excavators may have buckets, shears, or hammers attached.



(A) EXCAVATOR
BUCKET WITH TEETH



(B) DITCH-CLEANING EXCAVATOR
BUCKET WITHOUT TEETH



(C) EXCAVATOR WITH
REPLACEABLE BUCKET TEETH



(A) ROCK HAMMER



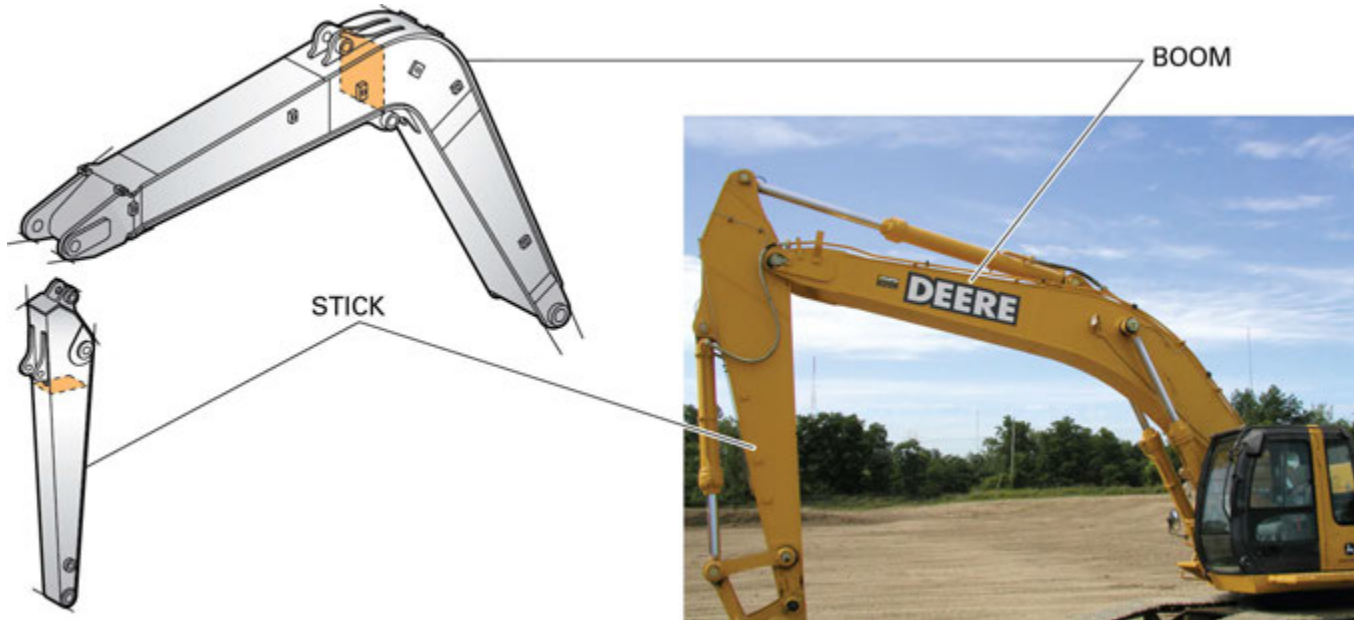
(B) BUCKET WITH THUMB



(C) SHEARS

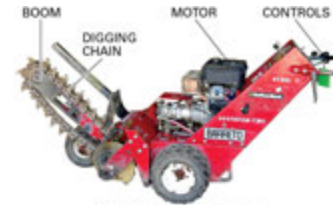
1.2.2 Excavators (3 of 3)

An excavating assembly consists of a boom and stick. Counterweight at the back of the excavator is used to balance the machine.



1.2.3 Trenchers

- Trenchers are machines used to dig trenches for the installation of underground services.
- The digging chain and boom are similar to the blade and chain of a chainsaw.



(A) PEDESTRIAN TRENCHER



(B) LARGE CONSTRUCTION TRENCHER



TRENCHER ATTACHMENT

(C) SKID STEER WITH TRENCHER ATTACHMENT

1.3.0 Mining Equipment

Mine Safety and Health Administration (MSHA) is the organization in charge of guarding the safety of miners to prevent death, disease, and injury resulting from mining work.



1.3.1 Aboveground Mining Equipment

- Aboveground mining involves digging down through the surface of the earth to reach particular minerals, such as coal, mineral ore, or stone.
- The equipment used for aboveground mining is usually large dozers, excavators, loaders, and rigid-frame trucks.



1.3.2 Belowground Mining Equipment

- To help reduce the risk of a collapse, the tunnel ceilings are kept low, which means that the machines are built low to the ground.
- Construction excavators and loaders may be used for higher-ceiling areas.
- Workers must pay attention to the air in these confined spaces and wear appropriate PPE.



WHEELED LOADER AND HYDRAULIC EXCAVATOR DIGGING A TUNNEL



UNDERGROUND MINING LOADER



UNDERGROUND TRUCK

Wrap Up – Trade Terms

Articulated: Two parts connected by a joint so as to move independently.

Haul road: A compacted dirt road used to move material and equipment on and off the site.

Pneumatic: Inflated with compressed air.

Ripper: A towed attachment with teeth used on dozers, motor graders and other machines to loosen heavily compacted soil and soft rock.

Scarifier: An attachment with teeth used on motor graders to loosen soil.



1.0.0 Section Review Question 1

1. The type of compactor used primarily for rolling paving material such as asphalt is the ____.
- a. steel wheel compactor
 - b. pneumatic tire compactor
 - c. vibratory compactor
 - d. sheepfoot compactor



1.0.0 Section Review Question 1 Answer

1. The type of compactor used primarily for rolling paving material such as asphalt is the ____.
- a. steel wheel compactor**
 - b. pneumatic tire compactor
 - c. vibratory compactor
 - d. sheepfoot compactor



1.0.0 Section Review Question 2

2. The machine that uses stabilizer legs is the _____.
- a. excavator
 - b. backhoe loader
 - c. scraper
 - d. motor grader



1.0.0 Section Review Question 2 Answer

2. The machine that uses stabilizer legs is the _____.
- a. excavator
 - b. backhoe loader**
 - c. scraper
 - d. motor grader



1.0.0 Section Review Question 3

3. The machines commonly used in underground mining are the loader and the _____.
a. backhoe
b. excavator
c. mining truck
d. scraper



1.0.0 Section Review Question 3 Answer

3. The machines commonly used in underground mining are the loader and the _____.
a. backhoe
b. excavator
c. mining truck
d. scraper



Next...

Section 2.0.0

Vehicle Drive System



*Read Sections 2.0.0 through 2.2.6 and
complete the Section Review.*